DATA ANALYSIS EXERCISE EXERCISE 4

( SQL Practice Exercise: UNION, Joins, Filtering, and Aggregates

Question 1: SQL Joins

1. Retrieve all employees and their assigned projects using an Inner Join.
   - Return: EmployeeID, FirstName, LastName, Department, Salary, ProjectID, ProjectName, Budget, Status.

SQL Query

```
SELECT employeeId,
        FirstName,
        lastName,
        Department,
        Salary,
        ProjectID,
        ProjectName,
        Budget,
        Status,
    FROM Employees
    INNER JOIN Projects
        ON EmployeeID = EmployeeID;
```

SQL Query Result Output

| EmployeID | FirstName | LastName | Department | Salary | Project ID | ProjectName | Budget | Status |
|---|---|---|---|---|---|---|---|---|
| 1 | John | Doe | IT | 70000 | 101 | AI Development | 100000 | Complete |
| 2 | John | Doe | IT | 70000 | 103 | Cybersecurity Audit | 75000 | Pending |
| 3 | Alice | Smith | HR | 60000 | 102 | Employee Training | 50000 | Ongoing |
| 4 | Bob | Johnson | Finance | 75000 | 104 | Financial Analysis | 90000 | Ongoing |
| 5 | Emma | Wilson | Sales | 65000 | 105 | Market Expansion | 65000 | Complete |
| 6 | Michael | Clark | Finance | 80000 | 106 | Risk Management | 80000 | Pending |

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2. Retrieve all employees and their assigned projects, including employees who have no projects using a LEFT JOIN.

- Return: EmployeeID, FirstName, LastName, Department, Salary, ProjectID, ProjectName, Budget, Status.

SQL Query

```
SELECT EmployeeID,
       FirstName,
       LastName,
       Department,
       ProjectName,
       Budget,
       Status,
  FROM Employees
 LEFT JOIN Projects
       ON EmployeeID = EmployeeID;
```

SQL Query Results Output

| EmployeeID | FirstName | LastName | Department | Salary | ProjectID | ProjectName | Budget | Status |
|---|---|---|---|---|---|---|---|---|
| 1 | John | Doe | IT | 70000 | 101 | AI Development | 100000 | Completed |
| 1 | John | Doe | IT | 70000 | 103 | Cybersecurity Audit | 75000 | Pending |
| 2 | Alice | Smith | HR | 60000 | 102 | Employee Training | 50000 | Ongoing |
| 3 | Bob | Johnson | Finance | 75000 | 104 | Financial Analysis | 90000 | Ongoing |
| 4 | David | Brown | IT | 72000 | NULL | NULL | NULL | NULL |
| 5 | Emma | Wilson | Sales | 65000 | 105 | Market Expansion | 65000 | Completed |
| 6 | Michael | Clark | Finance | 80000 | 106 | Risk Management | 80000 | Pending |

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3. Retrieve all Projects and their assigned employees, including projects that have no employees using a RIGHT JOIN.

- Return: ProjectID, ProjectName, Budget, Status, EmployeeID, FirstName, LastName, Department, Salary

SQL Query

```
SELECT ProjectID,
       ProjectName,
       Budget,
       Status,
       EmployeeID,
       FirstName,
       Department,
       Salary
FROM Employees
RIGHT JOIN Projects
    ON EmployeeID = EmployeeID;
```

SQL Query Result Output

| ProjectID | ProjectName | Budget | Status | EmployeeID | FirstName | Lastname | Department | Salary |
|---|---|---|---|---|---|---|---|---|
| 101 | AI Development | 100000 | Completed | 1 | John | Doe | IT | 70000 |
| 102 | Employee Training | 50000 | Ongoing | 2 | Alice | Smith | HR | 60000 |
| 103 | Cybersecurity Audit | 75000 | Pending | 1 | John | Doe | IT | 70000 |
| 104 | Financial Analysis | 90000 | Ongoing | 3 | Bob | Johnson | Finance | 75000 |
| 105 | Market Expansion | 65000 | Completed | 5 | Emma | Wilson | Sales | 65000 |
| 106 | Risk Management | 80000 | Pending | 6 | Michael | Clark | Finance | 80000 |

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4. Retrieve all employees and projects, including those without a match in their table using a FULL OUTER JOIN.

- Return: EmployeeID, FirstName, LastName, Department, Salary, ProjectID, ProjectName, Budget, Status

SQL Query

```
SELECT EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary,
       ProjectName,
       Budget,
       Status
  FROM Employees
  FULL OUTER JOIN Projects
    ON EmployeeID = EmployeeID;
```

SQL Query Output Results

| EmployeeID | FirstName | LastName | Department | Salary | Project ID | ProjectName | Budget | Status |
|---|---|---|---|---|---|---|---|---|
| 1 | John | Doe | IT | 70000 | 101 | AI Development | 100000 | Completed |
| 1 | John | Doe | IT | 70000 | 103 | Cybersecurity Audit | 75000 | Pending |
| 2 | Alice | Smith | HR | 60000 | 102 | Employee Training | 50000 | Ongoing |
| 3 | Bob | Johnson | Finance | 75000 | 104 | Financial Analysis | 90000 | Ongoing |
| 4 | David | Brown | IT | 72000 | NULL | NULL | NULL | NULL |
| 5 | Emma | Wilson | Sales | 65000 | 105 | Market Expansion | 65000 | Completed |
| 6 | Michael | Clark | Finance | 80000 | 106 | Risk Management | 80000 | Pending |

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Question 3: Filtering Statement

7. Retrieve employees who earn more than 70,000.

- Return: EmployeID, FirstName, LastName, Department, Salary.

SQL Query

```
SELECT EmployeeId,
       FirstName,
       LastName,
       Department,
       Salary,
FROM Employees
WHERE Salary > 70000;
```

SQL Query Output Results

| EmployeeID | FirstName | LastName | Department | Salary |
|---|---|---|---|---|
| 3 | Bob | Johnson | Finance | 75000 |
| 4 | David | Brown | IT | 72000 |
| 6 | Michael | Clark | Finance | 80000 |

8. Retrieve employees working in either IT or Finance departments.

- Return: EmployeeID, FirstName, LastName

SQL Query

```
SELECT EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary,
FROM Employees
WHERE Department IN ('IT', 'Finance');
```

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SQL Query Output Results

| EmployeeID | FirstName | LastName | Department | Salary |
|---|---|---|---|---|
| 1 | John | Doe | IT | 70000 |
| 3 | Bob | Johnson | Finance | 75000 |
| 4 | David | Brown | IT | 72000 |
| 6 | Michael | Clark | Finance | 80000 |

8. Retrieve employees working in either IT or Finance departments

- Return: EmployeeID, FirstName, LastName, Department, Salary.

SQL Query

```
SELECT EmployeeId,
       FirstName,
       LastName,
       Department,
       Salary,
  FROM Employees
  WHERE Department IN (
```

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9. Retrieve projects that are not yet completed

- Return: ProjectID, ProjectName, Budget, Status.

SQL Query

```
SELECT ProjectID,
       ProjectName,
       Budget,
       Status
FROM Projects
WHERE Status <> 'Completed';
```

SQL Query Output Results

| ProjectID | ProjectName | Budget | Status |
|---|---|---|---|
| 102 | Employee Training | 50000 | Ongoing |
| 103 | Cybersecurity Audit | 75000 | Pending |
| 104 | Financial Analysis | 90000 | Ongoing |
| 106 | Risk Management | 80000 | Pending |

10. Retrieve projects that have a budget greater than 70,000 and are not completed.

- Return: ProjectID, ProjectName, Budget, Status.

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